GippsAero GA8 Airvan (GA8)

Lycoming IO-540-K1A5, 300 hp · 8 seats · Typical GA8 (non-turbo) AFM procedures; fuel-injected engine, constant-speed propeller, cowl flaps, electric flaps 0°/14°/38°, LEFT/RIGHT/OFF fuel selector (no BOTH), left-rear sliding cargo door. Parachuting (door-open) operations only per the applicable AFM supplement.

V-speeds & limits

Vr	**60–65 KIAS (flaps 14°)**
Vx	**65 KIAS**
Vy	**78 KIAS**
Best glide	**75 KIAS (approx, 4,000 lb)**
Vs / Vso	**approx 57 / 50 KIAS (verify in AFM)**
Vfe	**100 KIAS (14°) / 86 KIAS (38°)**
Va	**119 KIAS (4,000 lb) / 103 KIAS (3,000 lb)**
Vno	**130 KIAS**
Vne	**170 KIAS**
Approach	**80–85 KIAS flaps up / 70–75 KIAS flaps 38° / 65 KIAS short field**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (stall strips present & undamaged, pitot cover off, sliding door track clear)**
Fuel quantity	**CHECK visually both tanks (approx 322 L / 85 US gal usable total)**
Oil quantity	**CHECK (12 qt max; add if below 9 qt)**
Cargo / baggage	**SECURED, tie-downs checked, weight & balance within limits**
Passenger briefing	**COMPLETE (belts, exits – crew door and sliding door, door handles, sterile cockpit)**
Sliding cargo door / crew doors	**CLOSED & LATCHED (handle fully down, door flush, DOOR annunciator out if fitted)**
Seats, belts, harnesses	**ADJUST & LOCK**
Brakes	**TEST & SET**
Control lock	**REMOVE**
Circuit breakers	**CHECK IN**
Electrical equipment / autopilot	**OFF**
Avionics master	**OFF**
Fuel selector	**LEFT or RIGHT (fullest tank)**
Cowl flaps	**OPEN**
Master switch (BAT & ALT)	**ON**
Annunciator panel	**TEST / CHECK**

Engine Start

Throttle	**OPEN approx 1/4 inch**
Propeller	**HIGH RPM (full forward)**
Mixture	**IDLE CUTOFF**
Beacon	**ON**
Propeller area	**CLEAR**
Fuel boost pump	**ON**
Mixture	**RICH until fuel flow indicates (approx 3–5 s), then IDLE CUTOFF**
Fuel boost pump	**OFF**
Ignition switch	**START (release when engine fires; max 10 s cranking)**
Mixture	**ADVANCE smoothly to RICH when engine fires**
Throttle	**ADJUST 1,000 RPM**
Oil pressure	**CHECK rising within 30 s (60 s cold) – if not, SHUT DOWN**
Alternator / voltmeter	**CHECK charging (approx 28 V)**
Avionics master	**ON**

Before Taxi

Altimeter	**SET**
Radios / transponder / GPS	**SET**
Flaps	**CHECK operation 0°–14°–38°–0° (indicator agrees)**
Parking brake	**RELEASE**
Brakes	**CHECK during initial taxi**
Flight instruments	**CHECK during turns (DG / HSI, turn coordinator, compass)**

Before Takeoff

Parking brake	**SET**
Seats, belts, harnesses	**CHECK SECURE**
Sliding cargo door / crew doors	**CLOSED & LATCHED**
Flight controls	**FREE & CORRECT**
Fuel quantity	**CHECK both tanks**
Fuel selector	**FULLEST TANK (do not change tanks just before takeoff)**
Mixture	**RICH (lean for field elevation above approx 3,000 ft)**
Elevator / rudder trim	**TAKEOFF**
Cowl flaps	**OPEN**
Throttle	**2,000 RPM**
Magnetos	**CHECK L / R / BOTH (max drop 175 RPM, max 50 RPM differential)**
Propeller	**CYCLE high to low RPM 2–3 times (RPM drop approx 300–500, oil pressure recovers)**
Engine instruments	**CHECK (oil press 55–95 psi, oil temp & CHT green, fuel flow steady)**
Vacuum / suction	**CHECK green (approx 4.5–5.5 in Hg)**
Alternator / annunciators	**CHECK (none lit)**
Throttle	**1,000 RPM**
Fuel boost pump	**ON**
Radios / transponder	**SET / ALT**
Strobe / landing lights	**ON**
Wing flaps	**14° (normal and short field takeoff)**
Takeoff briefing	**COMPLETE (rotate 60–65 KIAS, Vx 65 / Vy 78, abort and EFATO plan)**

After Takeoff / Climb

Airspeed	**78 KIAS (Vy) / 65 KIAS (Vx, obstacles); cruise climb 85–90 KIAS**
Wing flaps	**UP (above 70 KIAS, clear of obstacles)**
Fuel boost pump	**OFF (above approx 1,000 ft AGL; check fuel flow steady)**
Power	**FULL THROTTLE / 2,700 RPM to 1,000 ft, then 25 in Hg / 2,500 RPM**
Mixture	**RICH (lean above approx 3,000 ft per fuel flow placard / EGT)**
Cowl flaps	**OPEN**
Engine instruments	**CHECK (CHT / oil temp – lower nose if hot)**

Cruise

Power	**SET (typical 2,400 RPM / 22–24 in Hg, approx 75%)**
Mixture	**LEAN per fuel flow / EGT (approx 60 L/h – 16 GPH at 75%)**
Cowl flaps	**CLOSED / AS REQUIRED for CHT**
Fuel selector	**CHANGE TANKS at intervals (approx every 30–60 min) to keep balance**
Fuel boost pump	**ON while changing tanks, then OFF**
Fuel quantity / endurance	**CHECK against plan**
Sliding cargo door	**CLOSED & LATCHED (open in flight only for parachuting per supplement, within its airspeed limit)**

Descent

Altimeter	**SET**
Power	**REDUCE gradually (avoid rapid cooling – approx 1–2 in Hg per minute)**
Mixture	**ENRICH progressively**
Cowl flaps	**CLOSED**
Approach briefing	**COMPLETE**

Before Landing

Seats, belts, harnesses	**SECURE**
Fuel selector	**FULLEST TANK**
Fuel boost pump	**ON**
Mixture	**RICH**
Propeller	**HIGH RPM (full forward)**
Cowl flaps	**CLOSED (OPEN if go-around likely in hot conditions)**
Landing lights	**ON**
Autopilot	**OFF**
Airspeed	**80–85 KIAS (flaps up)**
Wing flaps	**14° below 100 KIAS / 38° below 86 KIAS**
Airspeed	**70–75 KIAS (flaps 38°) / 65 KIAS short field**
Go-around	**FULL POWER, flaps 14° immediately, 38°–14° only above 65 KIAS, flaps UP above 70 KIAS**

After Landing

Wing flaps	**UP**
Cowl flaps	**OPEN**
Fuel boost pump	**OFF**
Strobe / landing lights	**OFF / AS REQUIRED**

Shutdown

Parking brake	**SET**
Electrical equipment / autopilot / pitot heat	**OFF**
Avionics master	**OFF**
Mixture	**IDLE CUTOFF (pull full out)**
Ignition switch	**OFF (key removed)**
Master switch (BAT & ALT)	**OFF**
Fuel selector	**LEFT or RIGHT (as per operator; OFF for extended parking)**
Control lock	**INSTALL**
Sliding cargo door	**OPEN with care – hold door until fully back on track stop; CLOSE & LATCH when leaving aircraft**

Abnormal procedures

Alternator Failure / Low Voltage

ALT or LOW VOLT annunciator, ammeter discharge or voltmeter below approx 24 V

Alternator circuit breaker / field breaker	**CHECK IN**
Master switch (ALT half)	**OFF, then ON**
Ammeter / voltmeter	**CHECK**

▶ Charging restored?

Yes:

Ammeter / voltmeter	**MONITOR**

No:

Master switch (ALT half)	**OFF**
Non-essential electrical equipment	**OFF (lights, autopilot, second radio, pitot heat if not needed)**
Wing flaps	**PLAN early extension while battery power remains**

Deferred: Alternator failed – battery only, limited endurance (approx 30 min). Land as soon as practical; set flaps early, expect loss of fuel gauges and boost p

Low Oil Pressure

Oil pressure below 55 psi green arc or OIL PRESS annunciator

Oil pressure / oil temperature	**CHECK**

▶ Oil temperature also rising?

Yes:

Power	**REDUCE to minimum required**
Propeller	**EXPECT governor loss – RPM may rise to high-RPM stop**
Engine failure	**EXPECT – plan a precautionary landing**
Deferred: Low oil pressure with high temperature – land as soon as possible, expect engine failure.	**NOTED**

No – temp normal:

Flight	**CONTINUE to nearest suitable airport, monitor**
Deferred: Low oil pressure indication – land as soon as practical, maintenance check before further flight.	**NOTED**

High Oil Temperature / CHT

Oil temperature or CHT approaching red line

Cowl flaps	**OPEN**
Mixture	**RICH**
Power	**REDUCE**
Airspeed	**INCREASE (lower nose, 90–100 KIAS in climb)**
Oil pressure	**MONITOR**

▶ Temperature returning to green arc?

Yes:

Flight	**CONTINUE, monitor**

No:

Deferred: High engine temperature – land as soon as practical, minimum power, cowl flaps open.	**NOTED**

Fuel Flow Fluctuation / Rough Running

Fuel flow fluctuating, rough engine or power loss

Fuel boost pump ..... **ON**

Mixture ..... **ADJUST for smooth operation**

Fuel selector ..... **CHANGE TANK (check quantity)**

▶ Running smooth?

Yes:

Fuel boost pump ..... **OFF when fuel flow stable (ON again if fluctuation returns); check for fuel imbalance / tank venting**

No:

Magnetos ..... **CHECK L / R / BOTH, select smoothest**

Alternate induction air ..... **SELECT (if fitted) – suspect induction icing or filter blockage**

Deferred: Rough running engine – land as soon as practical, boost pump on, be ready for engine failure. ..... **NOTED**

Propeller Overspeed

RPM above 2,700 and not responding to propeller control

Throttle ..... **RETARD**

Oil pressure ..... **CHECK**

Propeller control ..... **DECREASE RPM (full aft; reset if governor responds)**

Airspeed ..... **REDUCE**

Throttle ..... **AS REQUIRED to keep RPM below 2,700**

Deferred: Propeller governor failed – RPM controlled with throttle and airspeed only; land as soon as practical. ..... **NOTED**

Flap Failure

Flaps do not move or indicator disagrees with switch

Flap circuit breaker ..... **CHECK / RESET once**

Flap switch ..... **RECYCLE**

▶ Flaps operable?

Yes:

Flap setting ..... **AS REQUIRED**

No – stuck up:

Approach ..... **FLAPS UP – 80–85 KIAS, flatter approach, longer runway**

Deferred: Flaps inoperative – flaps-up landing at 80–85 KIAS, increased landing distance. ..... **NOTED**

No – stuck down:

Airspeed ..... **KEEP BELOW Vfe for the position (100 KIAS at 14°, 86 KIAS at 38°)**

Fuel ..... **CHECK – higher burn at low speed, divert if needed**

Deferred: Flaps stuck extended – speed limited to Vfe, reduced range; land as soon as practical. ..... **NOTED**

Door Open / Unlatched in Flight

Crew door or sliding cargo door unlatched (DOOR annunciator if fitted)

Airspeed ..... **REDUCE (approx 80–90 KIAS)**

Aircraft ..... **CONTROL – open door has little effect on handling**

▶ Which door?

Crew door:

Window / vents ..... **OPEN (equalise pressure), then door CLOSE & LATCH**

Sliding cargo door:

Passengers ..... **SEATED, belts tight, away from door**

Door ..... **DO NOT ATTEMPT to close unless an approved door-open (parachuting) configuration applies and a crew member is harnessed**

Loose articles ..... **SECURE**

Airspeed ..... **KEEP within the door-open supplement limit; land as soon as practical**

Deferred: Sliding door open / unlatched – reduced airspeed, land as soon as practical; passengers seated and belted. ..... **NOTED**

Vacuum / Gyro Failure

Suction gauge low or attitude / heading indicator flags

Suction gauge ..... **CHECK (approx 4.5–5.5 in Hg)**

Attitude / heading indicators ..... **DISREGARD – cross-check**

Turn coordinator, compass, altimeter, airspeed ..... **USE for partial-panel flight**

IMC ..... **AVOID / EXIT**

Deferred: Vacuum failed – attitude and heading indicators unreliable; VMC only. ..... **NOTED**

Inadvertent Icing Encounter

Pitot heat ..... **ON**

Exit icing conditions ..... **TURN BACK / CHANGE ALTITUDE**

Cabin heat / windshield defrost ..... **FULL ON**

Power ..... **INCREASE / propeller RPM INCREASE to minimise propeller ice**

Alternate induction air ..... **SELECT (if fitted)**

Airspeed on approach ..... **INCREASE (flaps up or 14° max with residual ice)**

Deferred: Residual airframe ice – flaps-up / 14° landing, approach 10–15 KIAS faster; stall strips less effective, expect higher stall speed. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

On the runway

Throttle (memory)	**IDLE**
Brakes (memory)	**APPLY**
Wing flaps (memory)	**UP**
Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Ignition switch	**OFF**
Master switch (BAT & ALT)	**OFF**

Engine Failure After Takeoff

Airborne, insufficient runway remaining

LAND IMMEDIATELY

Airspeed (memory)	**75 KIAS flaps up / 70 KIAS flaps 14° / 65 KIAS flaps 38°**
Mixture (memory)	**IDLE CUTOFF**
Fuel selector (memory)	**OFF**
Ignition switch (memory)	**OFF**
Wing flaps (memory)	**AS REQUIRED (38° recommended once landing assured)**
Master switch (BAT & ALT) (memory)	**OFF (after flaps set)**
Doors	**UNLATCH (crew door)**

Land straight ahead with only small heading changes. Do not attempt to turn back to the runway.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	**75 KIAS (best glide), flaps UP**
Propeller (memory)	**LOW RPM (full aft – reduces drag)**
Fuel selector (memory)	**CHANGE TANK (check quantity on both)**
Fuel boost pump (memory)	**ON**
Mixture (memory)	**RICH (if no restart: IDLE CUTOFF, then advance slowly)**
Ignition switch (memory)	**BOTH (START if propeller has stopped)**

▶ Engine restarted?

Yes:

Propeller	**HIGH RPM, then as required**
Power	**AS REQUIRED (minimum needed)**
Fuel boost pump	**LEAVE ON if fuel flow fluctuates, else OFF**
Engine instruments	**MONITOR**
Deferred: Engine restarted after in-flight failure – land as soon as possible; suspect fuel starvation (tank), vapour or ignition.	**NOTED**

No – forced landing:

Forced landing without engine power:

Field / wind	**SELECT & PLAN (into wind if possible)**
Mayday / 7700	**TRANSMIT (time permitting)**
Passengers	**BRIEF – brace, exits (crew door, sliding door after stop)**
Seats, belts, harnesses	**SECURE**
Airspeed	**75 KIAS flaps up / 65 KIAS flaps 38°**
Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Ignition switch	**OFF**
Wing flaps	**38° when landing assured**
Master switch (BAT & ALT)	**OFF (after flaps set)**
Crew door	**UNLATCH prior to touchdown**
Touchdown	**SLIGHTLY TAIL LOW, minimum speed**
Brakes	**APPLY HEAVILY**

Engine Fire During Start

Cranking (memory) ..... CONTINUE (draw flames into engine)

▶ **Engine started?**

Yes:

- Power ..... 1,800 RPM for a few minutes
- Engine ..... SHUT DOWN & INSPECT for damage

No:

- Throttle ..... FULL OPEN
- Mixture ..... IDLE CUTOFF
- Cranking ..... CONTINUE
- Fuel boost pump ..... OFF
- Fuel selector ..... OFF
- Ignition switch ..... OFF
- Master switch (BAT & ALT) ..... OFF
- Fire extinguisher ..... OBTAIN & USE
- Aircraft ..... EVACUATE, inspect for damage

Engine Fire in Flight

LAND IMMEDIATELY

Mixture (memory) ..... IDLE CUTOFF

Fuel selector (memory) ..... OFF

Fuel boost pump (memory) ..... OFF

Master switch (BAT & ALT) (memory) ..... OFF

Cabin heat / air (memory) ..... OFF (overhead vents open)

Cowl flaps (memory) ..... CLOSED

Airspeed (memory) ..... 100 KIAS (increase if fire persists until extinguished)

Do not attempt an engine restart.

Forced landing ..... EXECUTE (see Engine Failure in Flight)

Electrical Fire in Flight

LAND AS SOON AS POSSIBLE

Master switch (BAT & ALT) (memory) ..... OFF

Avionics master (memory) ..... OFF

Vents / cabin air / heat (memory) ..... CLOSED

Fire extinguisher ..... ACTIVATE

All other electrical switches (except ignition) ..... OFF

After discharging the extinguisher, ventilate the cabin.

▶ **Fire out and electrical power necessary?**

Yes:

- Master switch (BAT & ALT) ..... ON
- Circuit breakers ..... CHECK – do not reset a faulted breaker
- Avionics master ..... ON, then radios one at a time with delay, check for smoke
- Vents / cabin air ..... OPEN when fire is completely out
- Deferred: Electrical fire – faulted circuit isolated, affected equipment unavailable. ..... NOTED

No:

- Continue with master off. No electric flaps, radios, fuel gauges or boost pump – flaps-up landing.
- Deferred: Electrical power off – flaps-up landing (approach 80–85 KIAS), no radio, no fuel gauges or boost pump. ..... NOTED

Cabin Fire

LAND AS SOON AS POSSIBLE

Master switch (BAT & ALT) (memory) ..... OFF

Vents / cabin air / heat (memory) ..... CLOSED

Fire extinguisher ..... ACTIVATE

Cabin ..... VENTILATE after fire is out

Land ..... AS SOON AS POSSIBLE, inspect for damage

Emergency Descent

Throttle (memory)	**IDLE**
Propeller (memory)	**HIGH RPM (full forward – maximum drag)**
Mixture (memory)	**RICH**
Wing flaps (memory)	**UP**
Airspeed (memory)	**130 KIAS smooth air (Vno) / 103–119 KIAS turbulence (Va)**
Cowl flaps	**CLOSED (avoid shock cooling)**
Power	**CLEAR ENGINE periodically**

Alternative with structural damage suspected: flaps 38° and 86 KIAS maximum.

Inadvertent Stall / Spin Entry

Uncommanded wing drop or rotation; spins are prohibited

Throttle (memory)	**IDLE**
Ailerons (memory)	**NEUTRAL**
Rudder (memory)	**FULL OPPOSITE to rotation**
Control column (memory)	**FORWARD to break stall (hold until rotation stops)**
Rudder (memory)	**NEUTRALISE as rotation stops, then ease out of dive**
Wing flaps	**UP if extended (once below 86 / 100 KIAS is assured)**

Stall is gentle with buffet from the leading-edge stall strips; recover at the first sign of buffet or wing rock. Avoid full-flap, high-power, low-speed manoeuvring, especially with aft loading.

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.7 (build 849ad9ab) · 2026-11-07